

Prism — System Architecture

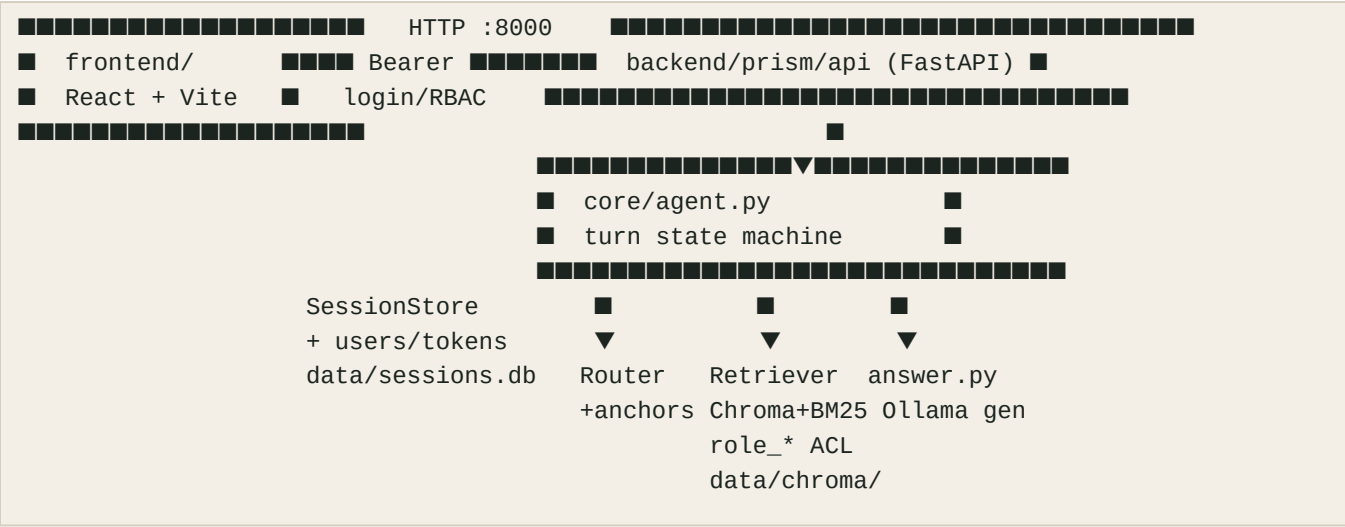
This document is the map of the repository: what Prism does, where each of the five knowledge bases comes from, how a chat turn runs, and how I plan to improve the system. Decision *rationale* (why Chroma, why Qwen, VRAM math) lives in [decisions.md](#). Prompt / non-LLM workflow inventory lives in [prompts.md](#). For a teaching walkthrough with worked examples (clarify, RBAC, policy math), see [system_guide.md](#) · [pdf/system_guide.pdf](#).

1. What Prism is

A **local-first RAG agent** for the Kohler–MITWPU Track 3 case study:

- One conversational agent over **five domains**: HR, Finance, Customer Support, Privacy, Legal
- One structured **Canonical Answer** per turn → re-render as prose / JSON / XML / Excel / email without re-asking the LLM
- **Ollama** for embeddings + generation (no cloud LLM)
- **Hand-rolled** Python turn control (`prism/core/agent.py`) — not LangGraph / LangChain agent loops

Target hardware: ~8GB NVIDIA VRAM (developed on RTX 5070 Laptop).



Layer	Role	Path
UI	Sessions, chat, format bar, Excel download, login	frontend/
API	Auth, sync chat, render, download	backend/prism/api/
Auth / RBAC	bcrypt users, Bearer tokens, denial log	backend/prism/core/auth.py, rbac.py

Layer	Role	Path
Agent	Ordered gates + RAG orchestration	backend/prism/core/agent.py
Index	One collection prism_kb + BM25 sidecar + role_* ACL	backend/data/chroma/
LLM	Gen / embed / titles	Ollama (OLLAMA_HOST)

2. The five knowledge bases

All domains land in **one** Chroma collection with domain metadata and boolean domain_<name> flags (so cross-tagged chunks, e.g. warranty = legal + CS, are filterable). Counts are approximate; rebuild reports the exact total (~497).

Domain	Nature	Source	Processed JSONL	~Chunks
Customer Support	Real	Crawl assist.kohler.com → data/raw/assist/	processed/customer_support.jsonl	~981
Privacy	Real	kohler.com Privacy Policy (rendered HTML cache under raw/kohler_legal/)	processed/privacy.jsonl	~91
Legal	Real	Assist warranties / Prop 65 / SDS / green-building + kohler.com Terms	legal_from_assist.jsonl + legal_from_kohler.jsonl	~451
HR	Synthetic	Authored data/synthetic/hr_policy.md (Meridian Fixtures Inc.)	processed/hr.jsonl	~49

Domain	Nature	Source	Processed JSONL	~Chunks
Finance	Synthetic	Authored data/synthetic/finance_policy.md	processed/finance.jsonl	

Ingest pipeline

1. `prism.ingest.crawl_assist` — sitemap → polite fetch (~1 req/s) → parse `__NEXT_DATA__` displayCard → classify CS vs legal cross-tags
2. `prism.ingest.parse_kohler_legal` — heading-boundary HTML chunks for privacy / T&C
3. `prism.ingest.build_synthetic` — HR / Finance markdown → JSONL
4. `prism.ingest.build_index` — merge all processed/*.jsonl → embed (nomic-embed-text) → rebuild Chroma + BM25 pickle

```
cd backend
uv run python -m prism.ingest.build_index
```

Path	Committed?	Notes
data/synthetic/*.md	Yes	Student-authored policies
data/processed/*.jsonl	Yes	Rebuild input
data/raw/	No (gitignored)	Crawl / HTML caches
data/chroma/	No	Local index + _embed_cache.jsonl + bm25_index.pkl
data/sessions.db	No	Chat state

3. Runtime: one chat turn

POST /api/sessions/{id}/chat → handle_turn → persist session → ChatResponse.

Gates run **in order**. Early returns skip the LLM when possible (safety + exact math).

Step	What	Module
0a	Jailbreak / prompt-exfil refuse	text_utils → agent

Step	What	Module
0b	CFO / fake policy-override refuse	text_utils → agent
0c	Deterministic leave / termination math	policy_math
1	Pure reformat of last_answer (JSON/XML/Excel/email)	agent + renderers/
2	Resolve clarification; vague cold-start clarify	agent
2c	“What did I ask N questions ago?” from history	agent
2d	Uploaded-doc gate — if the session has attachments and the user points at them (or upload retrieval is confident vs. the KBs), answer from the session-scoped prism_uploads collection	agent + uploads
3	Anaphora expansion (+ one small “focus” retrieval on the raw follow-up); long-range CL carry-forward email recall	agent
4	Route (embed + hit votes + anchors; sticky domain; ambiguity → ask)	router
5	Hybrid retrieve (dense + BM25 → RRF); multi-domain merge when needed	retriever + store
6	Relevance / fused confidence → else no_context_answer	retriever

Step	What	Module
7	One Ollama JSON generation → CanonicalAnswer (+ one repair)	answer
8	Polish (yes/no-only, anti-sycophancy, dedupe)	answer_polish
9	Render prose (or email); store last_answer / sources	renderers + memory

Formats later: POST .../render or chat-side pure reformat — both operate on the stored canonical object (no second retrieval).

Retrieval details

- **Dense:** Chroma cosine over nomic-embed-text vectors
- **Sparse:** BM25 over tokenized text; sidecar stores **ids + docs + metadatas** (no per-hit Chroma get)
- **Fusion:** Reciprocal rank fusion (RRF_K=60)
- **Confidence:** dense distance under domain floor **or** strong BM25 rank / dual-signal fused score (so exact Rs. / model tokens aren't false "no answer")
- **Multi-domain:** per-domain top_k=3 (≤4 domains), fuse ≤6; **skip agnostic pass** when ≥2 domain signals (or one signal with confident hits)

Deterministic guards (no LLM)

Documented also in prompts.md §4: jailbreak, CFO override, CL carry / leave-year math, yes/no-only collapse, session lookback, long-range carry-forward email, leading-number correction, fake-model refuse.

4. Key modules

Path	Role
prism/api/routes.py	Health (incl. Ollama + upload stats), sessions, chat, render, Excel, upload / uploads
prism/core/agent.py	Turn state machine
prism/core/router.py	Domain routing without an LLM
prism/core/retriever.py	Hybrid retrieve + confidence

Path	Role
prism/core/store.py	Chroma + BM25
prism/core/answer.py	CanonicalAnswer schema + generation
prism/core/answer_polish.py	Post-validation / constraints
prism/core/policy_math.py	Code-side leave arithmetic
prism/core/memory.py	Session state + SQLite (incl. uploaded_docs refs)
prism/core/uploads.py	Session-scoped upload store: parse → chunk → embed → prism_uploads collection; per-session BM25; TTL purge
prism/core/text_utils.py	Heuristics (jailbreak, multi-domain, formats)
prism/core/renderers/*	Prose / JSON / XML / Excel / email
prism/ingest/*	Crawl, parse, chunk, index
eval/stress/*	Live API adversarial harness

5. Eval & stress

- `eval/stress/harness.py` — black-box multi-turn cases against :8000
- Bank: `eval/stress/question_bank.py`
- Latest rollout: `backend/eval/results/stress/LATEST_REPORT.md`
- Must run **sequentially** on 8GB VRAM

Companion: `eval/run_eval.py`, `eval/bench_models.py`, `prism.ingest.validate`.

6. Configuration

Env var	Default	Purpose
PRISM_GEN_MODEL	qwen2.5:7b-instruct	Answer generation
PRISM_EMBED_MODEL	nomic-embed-text	Embeddings / routing

Env var	Default	Purpose
PRISM_TITLE_MODEL	qwen2.5:3b-instruct	Session titles
OLLAMA_HOST	http://localhost:11434	Ollama
PRISM_UPLOAD_MAX_BYTES	12 MB	Max upload size
PRISM_UPLOAD_TTL_HOURS	24	Purge uploaded vectors/files older than this (checked at startup)
PRISM_UPLOAD_KB_MARGIN	0.06	Implicit questions route to the upload only if within this dense-distance margin of the KB's best hit

Health: GET /api/health → status, chunks_indexed, configured models, ollama.{reachable,models_missing,ok}, and uploads.{chunks_indexed,ttl_hours,max_bytes,allowed_suffixes}.

7. Improvement roadmap

Ordered by impact × effort given **current** code (not generic RAG advice).

P0 — shipped

1. **Cache BM25 metadatas** in `bm25_index.pkl` — eliminate per-hit Chroma get
2. **Cap multi-domain cost** — skip agnostic retrieve when domain signals are clear
3. **Fused / lexical confidence** — dense floor **or** strong BM25 / dual RRF
4. **Health checks Ollama + models** — degraded when unreachable or models missing

P1 — shipped

5. **Deterministic Finance approval-band lookup** — claim amount → Section 2.1 band in `policy_math`
6. **SSE status streaming** — POST `.../chat/stream` emits stage events; final result is full ChatResponse (no partial JSON)
7. **Bypass-paraphrase stress** — `adv_06` / `adv_07` hit LLM+polish path (not regex short-circuits)
8. **Sticky→switch regression** — `switch_03` HR leave then faucet → `customer_support`; `switch_01` requires ≥ 2 domains

P2 / P3 — shipped

9. **UI states** — clarification / no-answer banners + confidence chips; turn metadata persisted

- 10. `scripts/rebuild_index.ps1` — fails loudly if required domain JSONL missing/empty
- 11. **Ollama keep-alive** — `PRISM_OLLAMA_KEEP_ALIVE` (default 25m) + startup warm
- 12. **Citation $\subseteq$ retrieved URLs** — polish drops sources not in retrieved chunks
- 13. **Multi-user session store** — SQLite WAL + RLock + busy timeout around session CRUD

Final sprint — shipped

- 14. **Session-scoped document upload (ad-hoc 6th domain)** — `POST .../upload`; PDF / DOCX / TXT / MD / CSV / JSON / HTML; separate `prism_uploads` collection filtered by `session_id`; dense + comparative-vs-KB confidence gate; purged on delete / remove / 24h TTL. Frontend: attach button, drag-and-drop, active-doc chips. See `decisions.md` §10.
- 15. **Anaphora focus retrieval** — extra `top_k=3` pass on the raw follow-up so topic pivots (“covered under warranty?”, “refund instead?”) surface the right chunks. See `decisions.md` §11.
- 16. **Water-conservation estimator** — EPA WaterSense / Fix-a-Leak liters/day note on leak/drip/running-toilet Support answers (`water_math.py`). See `decisions.md` §12.
- 17. **Landing page** — headline, six domain cards, four-step how-it-works, curated demo questions.

Track progress against this list in PRs; update this section when an item ships.

8. Related docs

Doc	Contents
decisions.md	Why each infra/model choice
prompts.md	Prompts + non-LLM workflows
demo_script.md	Demo narrative
deck.md	Slide outline
../backend/eval/stress/README.md	How to run stress